INCH-POUND

MIL-PRF-19500/737
13 November 2006

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, SCHOTTKY, DUAL, CENTER TAP,
TYPES 1N7039CCT1, 1N7039CCU1 AND 1N7047CCT3, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, Schottky dual power rectifier diodes for use in high frequency switching applications. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (TO-254AA), figure 2 (U1), and figure 3 (TO-257AA).

1.3 Maximum ratings. Unless otherwise specified, maximum ratings ($T_C$ = +25°C).

| Column 1 | Column 2 | Column 3 | Column 4 | Column 5 | Column 6 |
|---|---|---|---|---|---|
| Types | $V_{RWM}$ | $I_O$ (1)(2) $T_C$ = +100°C | $I_{FSM}$ (2) $t_p$ = 8.3 ms, $T_C$ = +25°C | $R_{\Theta JC}$ (2) (3) | $T_{STG}$ and $T_J$ |
| | V dc | A dc | A (pk) | °C/W | °C |
| 1N7039CCT1 | 150 | 35 | 180 | 1.9 | -65 to +150 |
| 1N7039CCU1 | 150 | 35 | 200 | 1.67 | |
| 1N7047CCT3 | 150 | 16 | 120 | 1.85 | |

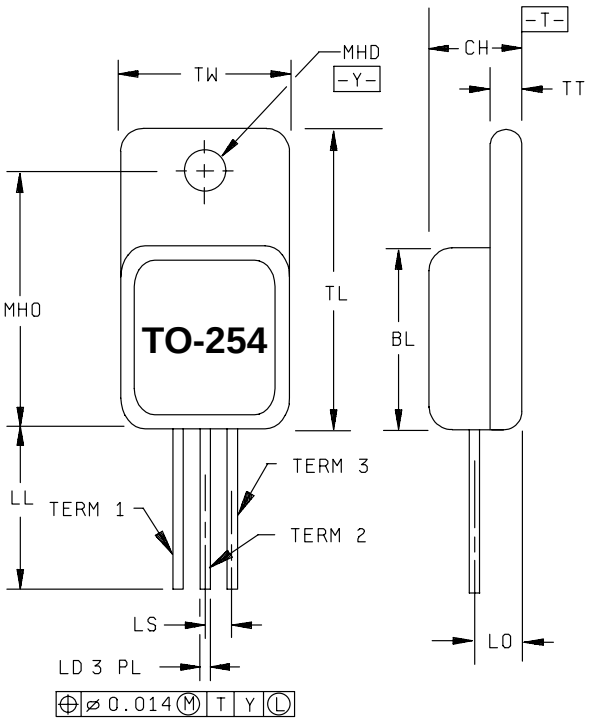
(1) See temperature-current derating curves in figures 4, 5, and 6.
(2) Each leg
(3) See figures 7, 8, and 9.

1.4 Primary electrical characteristics.
$R_{\Theta JC}$ = 0.95 °C/W maximum for entire package for 1N7039CCT1, $R_{\Theta JA}$ = 40°C/W maximum, per leg;
RΘJC = 0.83 °C/W maximum for entire package for 1N7039CCU1;
$R_{\Theta JC}$ = 0.95 °C/W maximum for entire package for 1N7047CCT3.

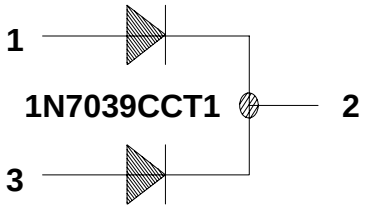
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AMSC N/A FSC 5961

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| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .535 | .545 | 13.59 | 13.84 |
| CH | .249 | .260 | 6.32 | 6.60 |
| LD | .035 | .045 | 0.89 | 1.43 |
| LL | .510 | .570 | 12.95 | 14.48 |
| LO | .150 BSC | | 3.81 BSC | |
| LS | .150 BSC | | 3.81 BSC | |
| MHD | .139 | .149 | 3.53 | 3.78 |
| MHO | .665 | .685 | 16.89 | 17.40 |
| TL | .790 | .800 | 20.07 | 20.32 |
| TT | .040 | .050 | 1.02 | 1.27 |
| TW | .535 | .545 | 13.59 | 13.84 |

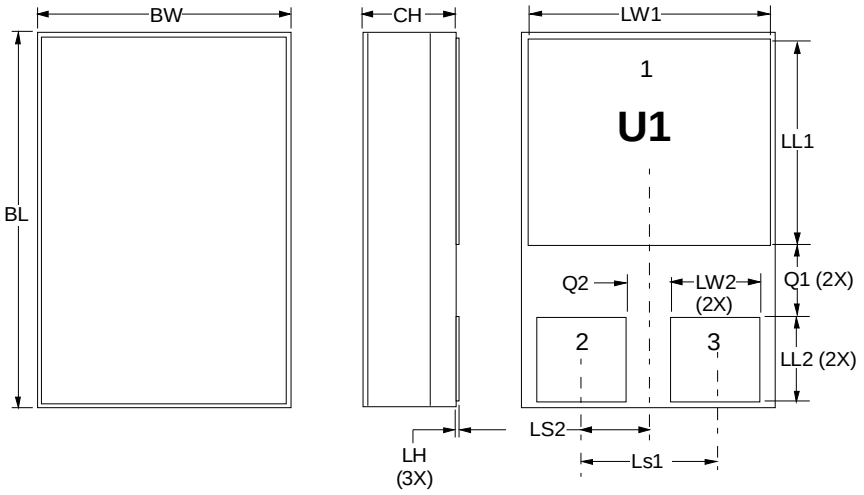



NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

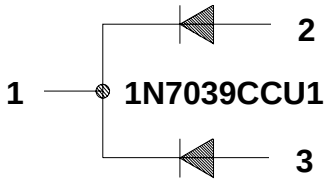
FIGURE 1. Dimensions and configuration for 1N7039CCT1 (TO-254AA).



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| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .620 | .630 | 15.75 | 16.00 |
| BW | .445 | .455 | 11.30 | 11.56 |
| CH | .129 | .139 | 3.28 | 3.53 |
| LH | .010 | .020 | 0.25 | 0.51 |
| LL1 | .410 | .420 | 10.41 | 10.67 |
| LL2 | .152 | .162 | 3.86 | 4.11 |
| LS1 | .200 | .220 | 5.08 | 5.59 |
| LS2 | .100 | .110 | 2.54 | 2.79 |
| LW1 | .370 | .380 | 9.40 | 9.65 |
| LW2 | .135 | .145 | 3.43 | 3.68 |
| Q1 | .035 | | 0.89 | |
| Q2 | .050 | | 1.27 | |
| Term 1 | Cathode | | | |
| Term 2 | Anode | | | |
| Term 3 | Anode | | | |

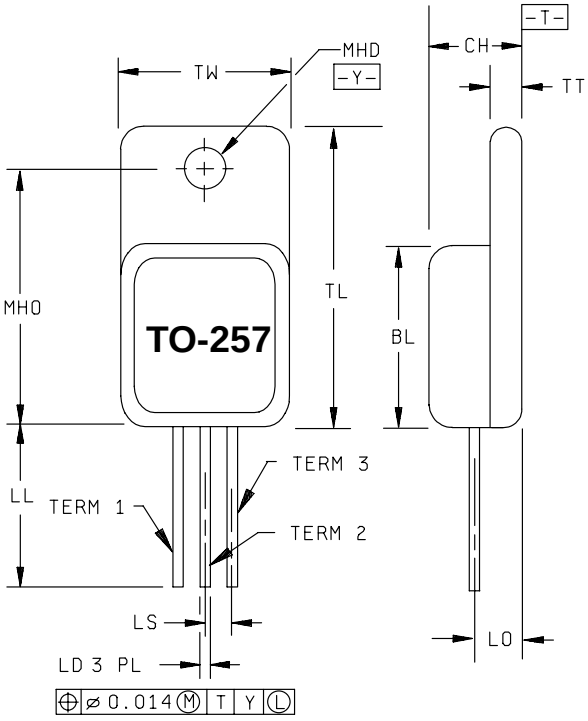



NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.

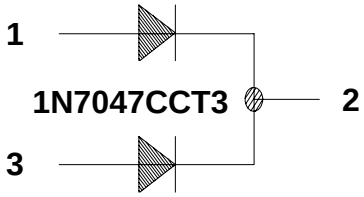
FIGURE 2. Dimensions and configuration, 1N7039CCU1.



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| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .410 | .430 | 10.41 | 10.92 |
| CH | .190 | .200 | 4.83 | 5.08 |
| LD | .025 | .040 | 0.64 | 1.02 |
| LL | .500 | .625 | 12.70 | 19.05 |
| LO | .120 BSC | | 3.05 BSC | |
| LS | .100 BSC | | 2.54 BSC | |
| MHD | .140 | .150 | 3.56 | 3.81 |
| MHO | .527 | .537 | 13.39 | 13.64 |
| TL | .645 | .665 | 16.38 | 16.89 |
| TT | .035 | .045 | 0.89 | 1.14 |
| TW | .410 | .420 | 10.41 | 10.67 |




NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 3. Dimensions and configuration for 1N7047CCT3.



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1, 2, and 3 herein. Methods used for electrical isolation of the terminal feedthroughs for the TO-254 or TO-257 packages shall employ materials that contain a minimum of 90 percent $Al_2O_3$ (ceramic).

3.4.1 Polarity. Polarity and terminal configuration shall be in accordance with figures 1 and 2 herein.

3.4.2 Lead finish and formation. Lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500 and herein. Where a choice of finish is desired, it shall be specified in the acquisition document (see 6.2). When lead formation is performed on TO-254 or TO-257 leads, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14 of MIL-PRF-19500 and 100 percent dc testing in accordance with table I, subgroup 2 herein.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in tables I and II herein.



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3.7 Marking. Marking shall be in accordance with MIL-PRF-19500 and herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II herein).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening (JANS, JANTXV, and JANTX levels). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table E-IV of MIL-PRF-19500) | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| (1)(2) 3b | Method 4066 of MIL-STD-750, condition A, one pulse, $I_O$ = 0, $V_{RWM}$ = 0, see 1.3 herein, column 4. | Method 4066 of MIL-STD-750, condition A, one pulse, $I_O$ = 0, $V_{RWM}$ = 0, see 1.3 herein, column 4. |
| (1) 3c | Thermal impedance (see 4.3.2) | Thermal impedance (see 4.3.2) |
| 3d | Peak reverse energy test (see 4.3.3) | Peak reverse energy test (see 4.3.3) |
| 9, 10 | Not applicable | Not applicable |
| 11 | $V_{F1}$ and $I_{R1}$ | $V_{F1}$ and $I_{R1}$ |
| 12 | See 4.3.1 | See 4.3.1 |
| 13 | Subgroup 2 and 3, of table I herein, $V_{F1}$ and $I_{R1}$, excluding thermal impedance; $\Delta V_{F1}$ ≤50 mV (pk); $\Delta I_{R1}$ = ± 100 percent from the initial value or 100μA dc, whichever is greater. | Subgroup 2, of table I herein excluding thermal impedance; $V_{F1}$ and $I_{R1}$; $\Delta V_{F1}$ ≤50 mV (pk); $\Delta I_{R1}$ = ± 100 percent from the initial value or 100μA dc, whichever is greater. |

(1) Thermal impedance and surge shall be performed any time after screen 3a and before screen 13.
(2) Surge shall precede thermal impedance.



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4.3.1 Power burn-in conditions. Burn-in conditions are as follows: Method 1038 of MIL-STD-750, test condition A. $T_C$ = +125°C; $V_R$ = 120 V dc.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{MD}$ and $V_C$. Measurement delay time ($t_{MD}$) = 70 μs max. See table III, group E, subgroup 4 herein.

4.3.3 Peak reverse energy test. The peak reverse energy test is to be performed using the circuit as shown on figure 10 or equivalent. The Schottky rectifier under test must be capable of absorbing the reverse energy, as follows: $I_{RM}$ = 170 mA, $V_{RSM}$ = 150 V minimum, L = 150 mH.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500, and table I herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables VIa (JANS) and VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2, forward voltage test ($V_{F1}$) and reverse leakage test ($I_{R1}$) herein. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Group B inspection, table E-VIa (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B4 | 1037 | $\Delta T_C$ = +85°C, $I_F$ = 2 A minimum for 2,000 cycles. |
| B5 | 1038 | Condition B, $I_F$ = 1 A dc minimum, adjust $T_A$ and $I_F$ to achieve $T_J$ = +150°C, +0° -10°C, t = 240 hours minimum; (heat sinking allowed). |
| B6 | 4081 | Limit for thermal resistance for 1N7039CCT1 is 1.90°C/W for each leg.<br>Limit for thermal resistance for 1N7039CCU1 is 1.67°C/W each leg.<br>Limit for thermal resistance for 1N7047CCT3 is 1.85°C/W for each leg. |

4.4.2.2 Group B inspection, table E-VIb (JANTXV, JANTX, JAN and) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1037 | $\Delta T_C$ = +85°C minimum, $I_F$ = 2 A minimum for 2,000 cycles. |
| B5 | 3101<br>or 4081 | Limit for thermal resistance for 1N7039CCT1 is 1.90°C/W for each leg.<br>Limit for thermal resistance for 1N7039CCU1 is 1.67°C/W each leg.<br>Limit for thermal resistance for 1N7047CCT3 is 1.85°C/W for each leg. |



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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2, forward voltage test ($V_{F1}$) and reverse leakage test ($I_{R1}$) herein. Delta measurements shall be in accordance with table II herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Condition A, weight = 10 lbs, t = 15 seconds. |
| C6 | 1037 | $\Delta T_C$ = +85°C, minimum, $I_F$ = 2 A minimum for 6,000 cycles. |

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-IX of MIL-PRF-19500, and table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein except $Z_{\Theta JX}$ need not be performed after group E subgroup 2.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



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TABLE I. Group A inspection.

| Inspection 1/ 2/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance | 3101 | See 4.3.2 | $Z_{\Theta JX}$ | | | |
| Forward voltage | 4011 | Pulsed test (see 4.5.1) | $V_{F1}$ | | | |
| 1N7039CCT1 | | $I_F$ = 15A (pk) | | | 1.13 | V dc |
| 1N7039CCU1 | | $I_F$ = 15A (pk) | | | 1.13 | V dc |
| 1N7047CCT3 | | $I_F$ = 8A (pk) | | | 0.91 | V dc |
| Forward voltage | 4011 | Pulsed test (see 4.5.1) | $V_{F2}$ | | | |
| 1N7039CCT1 | | $I_F$ = 35 A(pk) | | | 1.60 | V dc |
| 1N7039CCU1 | | $I_F$ = 35 A(pk) | | | 1.60 | V dc |
| 1N7047CCT3 | | $I_F$ = 16 A(pk) | | | 1.13 | V dc |
| Reverse current | 4016 | DC method | $I_{R1}$ | | | |
| 1N7039CCT1 | | $V_R$ = 150 V | | | 0.5 | mA dc |
| 1N7039CCU1 | | $V_R$ = 150 V | | | 0.5 | mA dc |
| 1N7047CCT3 | | $V_R$ = 150 V | | | 0.5 | mA dc |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_C$ = +125 °C | | | | |
| Forward voltage | | Pulsed test (see 4.5.1) | $V_{F3}$ | | | |
| 1N7039CCT1 | | $I_F$ = 15 A(pk) | | | 0.86 | V dc |
| 1N7039CCU1 | | $I_F$ = 15 A(pk) | | | 0.86 | V dc |
| 1N7047CCT3 | | $I_F$ = 8 A(pk) | | | 0.73 | V dc |
| Forward voltage | | Pulsed test (see 4.5.1) | $V_{F4}$ | | | |
| 1N7039CCT1 | | $I_F$ = 35 A(pk) | | | 1.20 | V dc |
| 1N7039CCU1 | | $I_F$ = 35 A(pk) | | | 1.20 | V dc |
| 1N7047CCT3 | | $I_F$ = 16 A(pk) | | | 0.94 | V dc |
| Reverse current | 4016 | DC method; | $I_{R2}$ | | | |
| 1N7039CCT1 | | $V_R$ = 150 V | | | 15 | mA dc |
| 1N7039CCU1 | | $V_R$ = 150 V | | | 15 | mA dc |
| 1N7047CCT3 | | $V_R$ = 150 V | | | 15 | mA dc |

See footnotes at end of table.



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TABLE I. Group A inspection – Continued.

| Inspection 1/ 2/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 - continued | | | | | | |
| Low temperature operation: | | $T_C$ = -55°C | | | | |
| Forward voltage | 4011 | Pulsed test (see 4.5.1) | $V_{F5}$ | | | |
| 1N7039CCT1<br>1N7039CCU1<br>1N7047CCT3 | | $I_F$ = 15 A dc<br>$I_F$ = 15A (pk)<br>$I_F$ = 8 A dc | | | 1.35<br>1.35<br>1.02 | V dc<br>V dc<br>V dc |
| Forward voltage | 4011 | Pulsed test (see 4.5.1) | $V_{F6}$ | | | |
| 1N7047CCT3 | | $I_F$ = 16 A dc | | | 1.18 | V dc |
| Subgroup 4 | | | | | | |
| Junction capacitance | 4001 | $V_R$ = 5 V dc, f = 1 MHz, $V_{SIG}$ = 50 mV (p-p) | $C_J$ | | 350 | pF |
| Subgroup 5 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 6 | | | | | | |
| Surge | 4066 | See column 4 of 1.3. ten surges each internal diode. 60 seconds between surges. (see 4.5.1) | | | | |
| Electrical measurements | | See table I, subgroup 2 herein. | | | | |
| Subgroup 7 | | | | | | |
| Dielectric withstanding voltage | 1016 | $V_R$ = 500 V dc; all leads shorted; measure from leads to case | DWV | | 10 | µA dc |
| Scope display evaluation | 4023 | Stable only | | | | |
| Electrical measurements | | See table I, subgroup 2 herein. | | | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ Electrical characteristics apply to all package styles and polarities.



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TABLE II. Groups B and C delta requirements. 1/ 2/ 3/ 4/ 5/ 6/

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Forward voltage | 4011 | | $\Delta V_{F1}$ | ±50 mV dc from initial reading. | | |
| | 1N7039CCT1 | | $I_F$ = 15 A (pk) pulsed (see 4.5.1) | | | | |
| | 1N7039CCU1 | | $I_F$ = 15 A (pk) pulsed (see 4.5.1) | | | | |
| | 1N7047CCT3 | | $I_F$ = 8 A (pk) pulsed (see 4.5.1) | | | | |
| 2. | Reverse current | 4016 | Vr = 150V | $\Delta I_{R1}$ | ≤ 100 percent from the initial value or 100μA dc, whichever is greater. | | |
| 3. | Thermal impedance | 3101 | See 4.3.2 | $Z_{\Theta JX}$ | | | |

1/ Each internal diode.
2/ The electrical measurements for table E-VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 4, see table II herein, steps 1, 2, and 3.
   b. Subgroup 5, see table II herein, steps 1 and 2.
3/ The electrical measurements for table E-VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table II herein, steps 1, 2, and 3.
   b. Subgroup 3, see table II herein, steps 1, 2, and 3.
   c. Subgroup 6, see table II herein, steps 1 and 2.
4/ The electrical measurements for table E-VII of MIL-PRF-19500 are as follows:
   a. Subgroups 2 and 3, see table II herein, steps 1, 2, and 3 for all levels.
   b. Subgroup 6, see table II herein, steps 1, 2, and 3 for all levels.
5/ Devices which exceed the table I limits for this test shall not be accepted.
6/ The electrical measurements for table E-IX of MIL-PRF-19500 are as follows:
   a. Subgroup 1, see table II herein, steps 1, 2, and 3.
   b. Subgroup 2, see table II herein, steps 1 and 2.



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TABLE III. Group E inspection (all quality levels) – for qualification and requalification only.

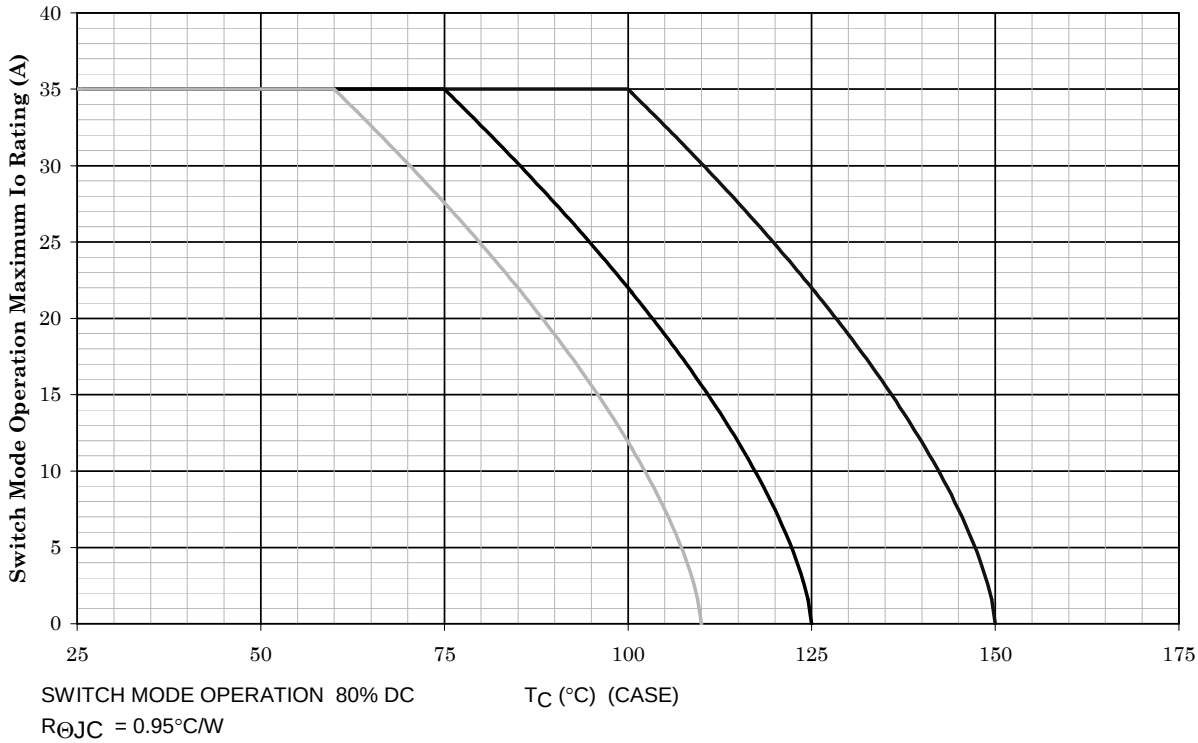
| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | n = 12, c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition G, 500 cycles, -55°C to +150°C. | |
| Hermetic seal | 1071 | | |
| Electrical measurements | | See table I, subgroup 2. | |
| Subgroup 2 | | | n = 12, c = 0 |
| Life test | 1048 | t = 1,000 hours, $T_C$ = +125°C, $V_R$ = 80 percent rated voltage (see 1.3, column 2 herein). | |
| Electrical measurements | | See table I subgroup 2. | |
| Subgroup 4 | | | |
| Thermal impedance curves | | See MIL-PRF-19500. | |
| Subgroup 6 | | | n = 3, c = 0 |
| ESD | 1020 | | |
| Subgroup 7 1/ | | | n = 5, c = 0 |
| Surge | 4066 | | |
| 1N7039CCT1 | | Condition A, $T_A$ = +25°C $I_{FSM}$ = 180 A, ten surges of 8.3 ms superimposed on $I_O$. $V_R$ = 0; $I_O$ = 10 A pk half sine wave, continuous. | |
| 1N7039CCU1 | | Condition A, $T_A$ = +25°C, $I_{FSM}$ = 200 A, ten surges of 8.3 ms superimposed on $I_O$. $V_R$ = 0; $I_O$ = 10 A pk half sine wave, continuous. | |
| 1N7047CCT3 | | Condition A, $T_A$ = +25°C $I_{FSM}$ = 120 A, ten surges of 8.3 ms superimposed on $I_O$. $V_R$ = 0; $I_O$ = 10 A pk half sine wave, continuous. | |
| Electrical measurements | | See table I subgroup 2 ($V_F$ and $I_R$ only). | |

1/ Each individual diode.



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**TEMPERATURE-CURRENT DERATING CURVE**
**1N7039CCT1**




NOTES:

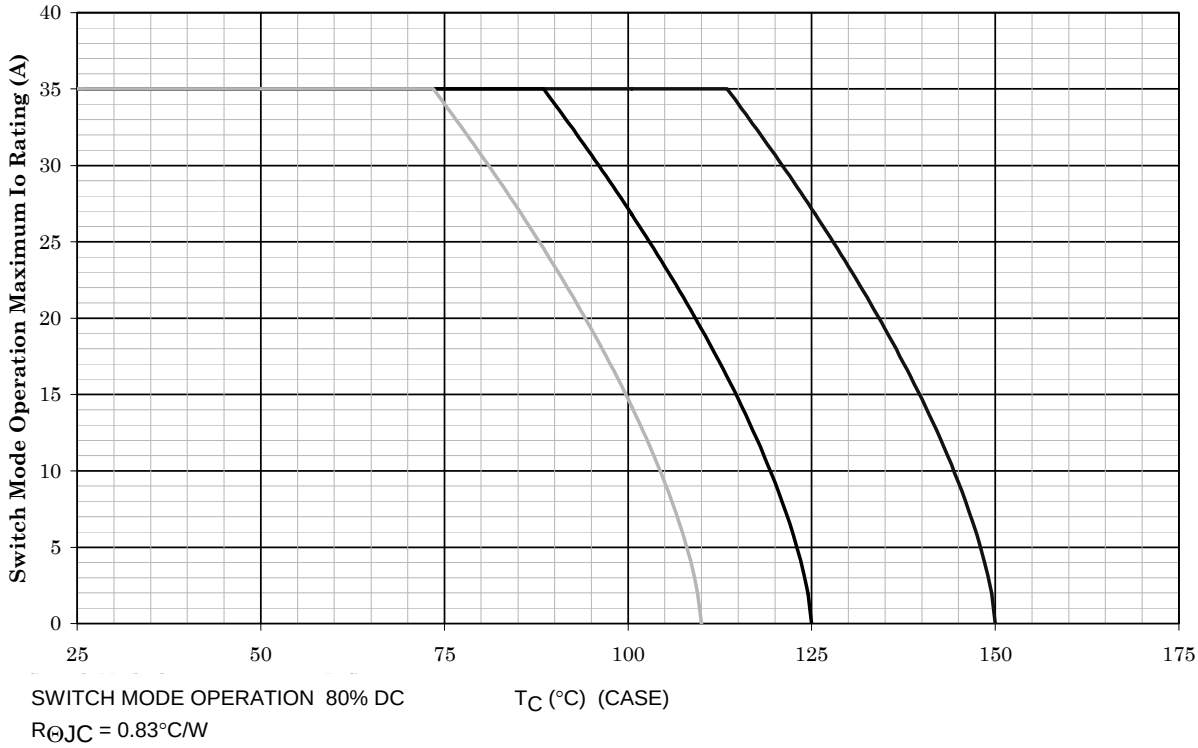
1. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate current for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 150°C$) and current rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 125°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq$, 125°C, and 110°C to show current rating where most users want to limit $T_J$ in their application.

FIGURE 4. Temperature-current derating curve (entire package) 1N7039CCT1.



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**TEMPERATURE-CURRENT DERATING CURVE**
**1N7039CCU1**




NOTES:

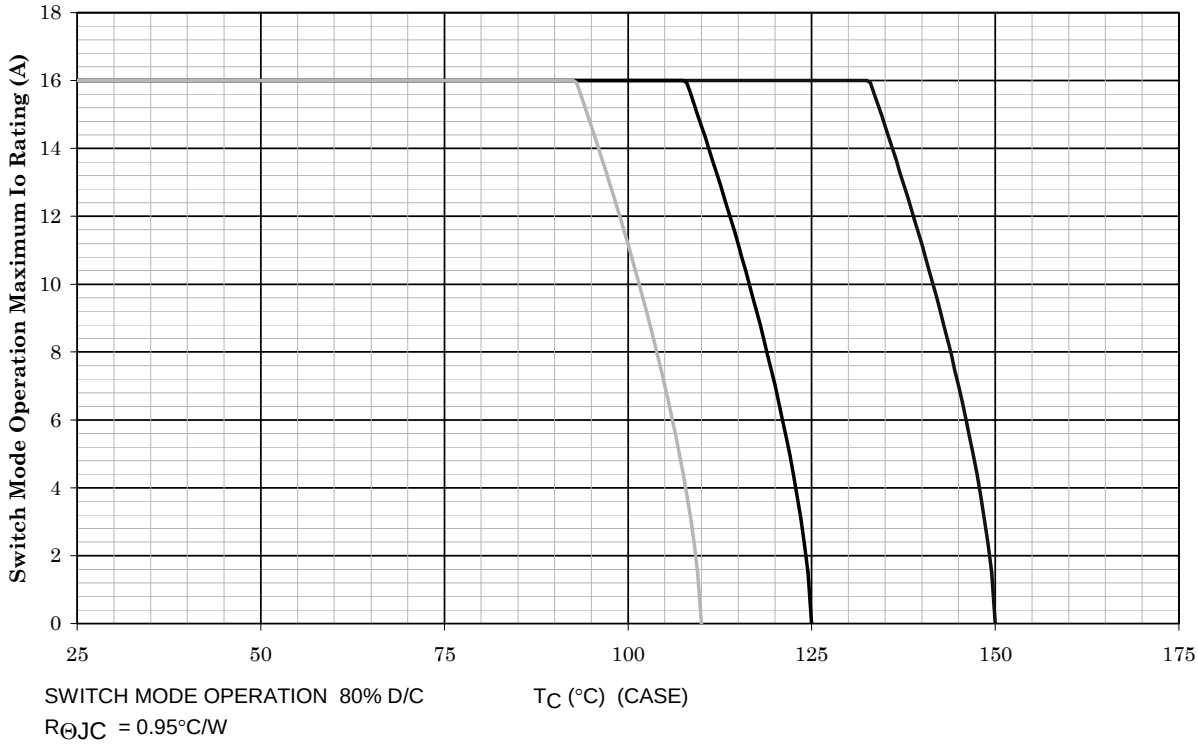
1. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate current for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 150°C) and current rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ 125°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤, 125°C, and 110°C to show current rating where most users want to limit $T_J$ in their application.

FIGURE 5. Temperature-current derating curve (each leg) for 1N7039CCU1.



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**TEMPERATURE-CURRENT DERATING CURVE**
**1N7047CCT3**




NOTES:

1. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate current for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 150°C) and current rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ 125°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤, 125°C, and 110°C to show current rating where most users want to limit $T_J$ in their application.

FIGURE 6. Temperature-current derating curve, 1N7047CCT3.



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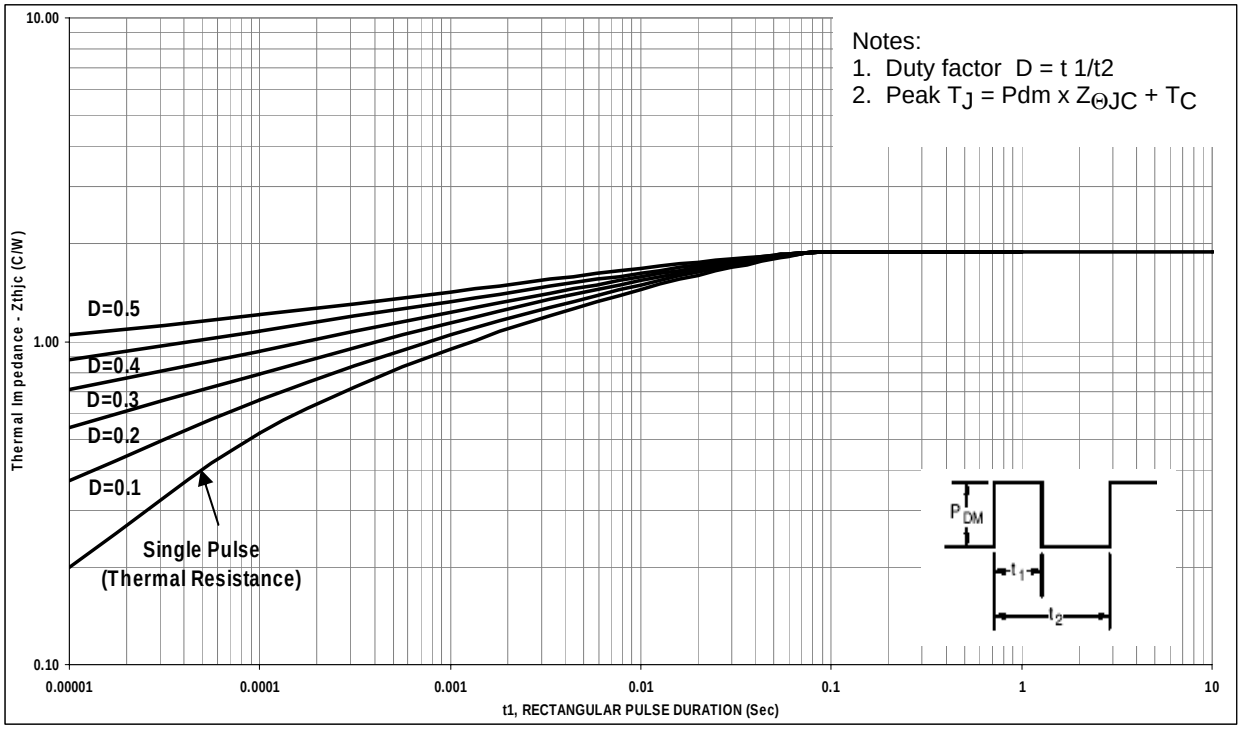



FIGURE 7. Thermal impedance (each leg) 1N7039CCT1.

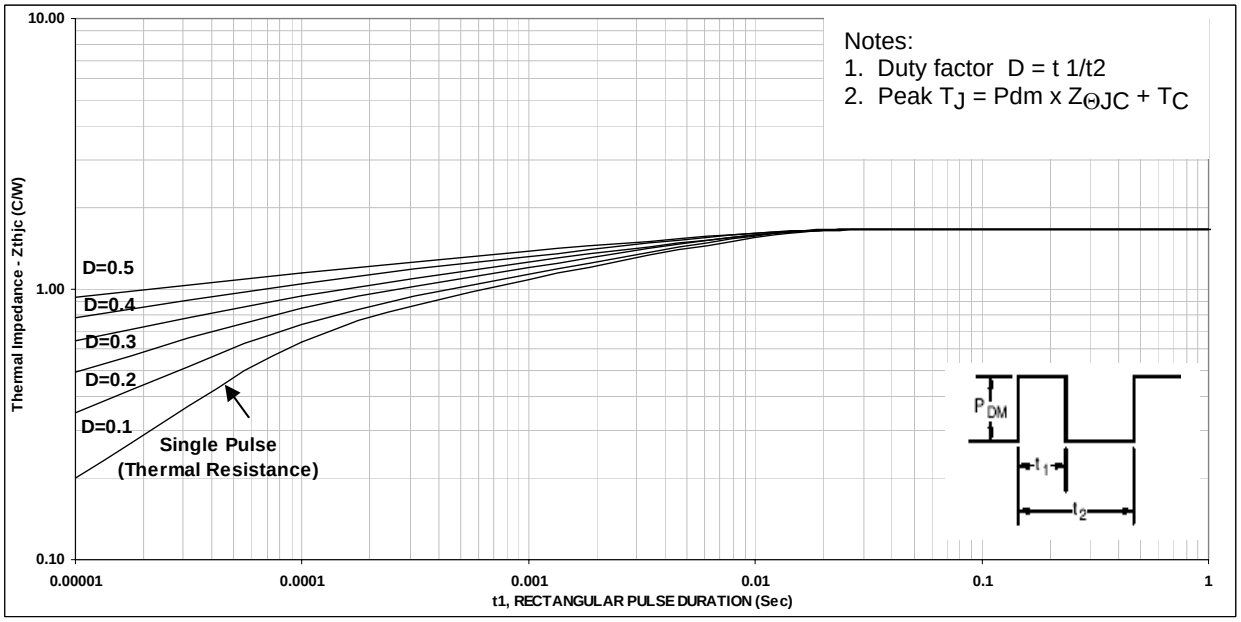



FIGURE 8. Thermal impedance (each leg), 1N7039CCU1.



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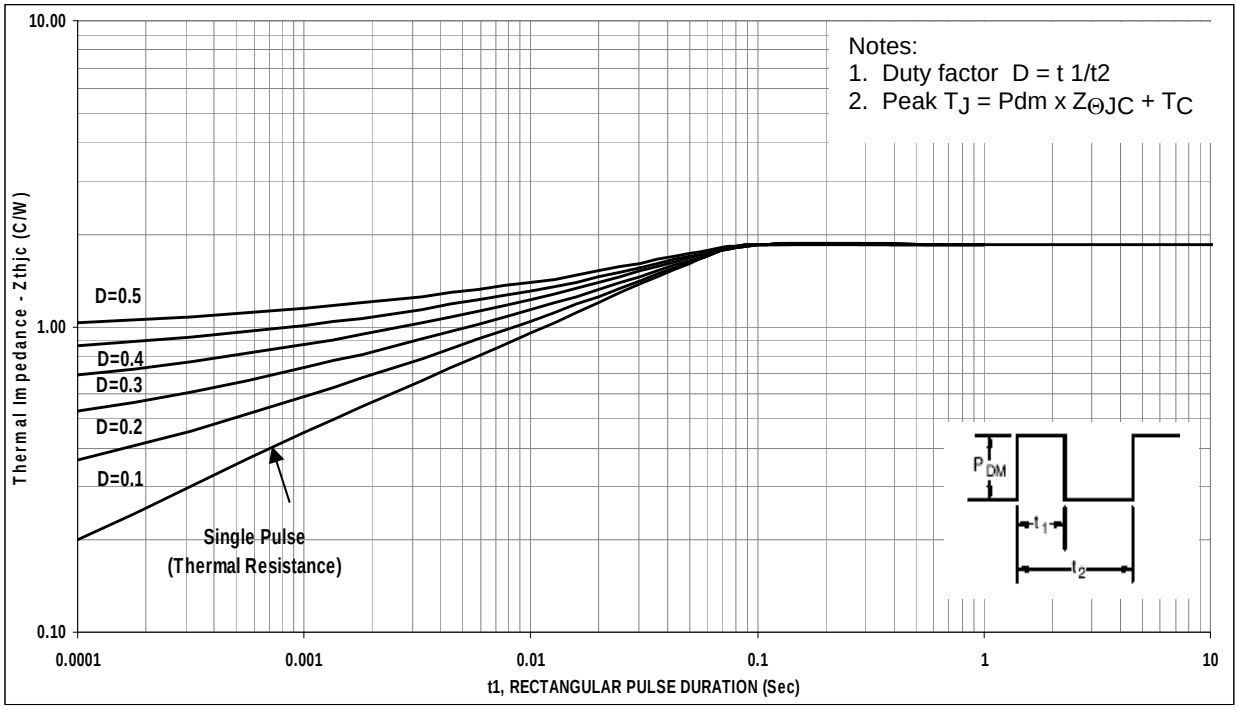
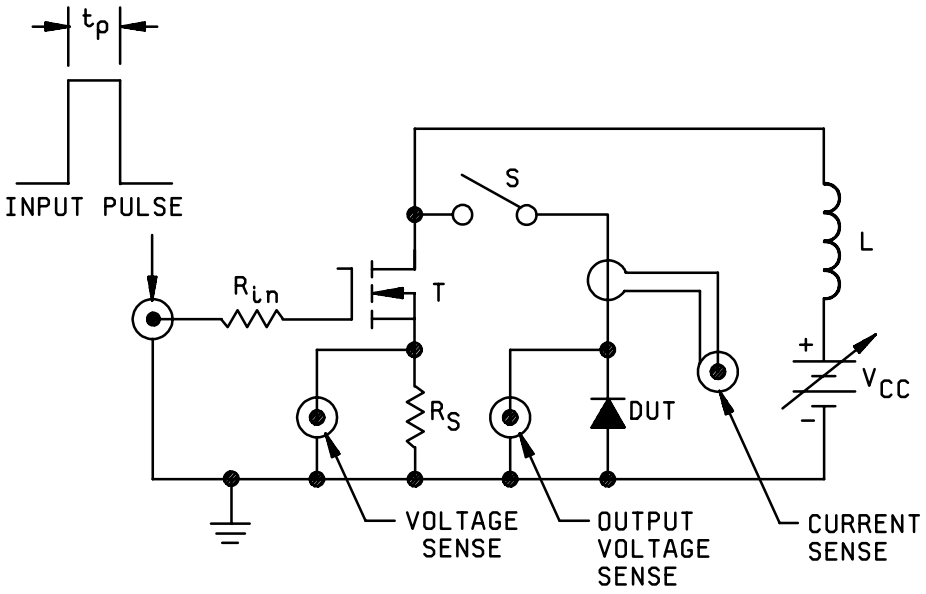



FIGURE 9. Thermal impedance, 1N7047CCT3 (each leg).



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Input pulse $R_{in}$ = 50 ohms, 1 watt
$V_G$ = 10 Volts, $R_S$ = 0.1 ohms, 1 watt
$Z_G$ = 50 ohms
L = 150mH
P.W. ≈ 30 μs
Duty cycle ≤ 1 percent, T = IRF350/2N6768 or equivalent

Procedure:
1. With S open, adjust pulse width to test current of 1 amp across $R_S$.
2. Close S, verify test current with current sense.
3. Read peak output voltage (see 4.3.3).

FIGURE 10. Peak reverse energy test circuit.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish and formation (see 3.4.2).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil .

6.4 Cross reference substitution list. A PIN for PIN replacement table follows, and these devices are directly interchangeable.

| Non-preferred PIN | Preferred PIN |
|---|---|
| 12CGQ150<br>12CLQ150<br>16CYQ150, 16CYQ150C | JANS, JANTXV, JANTX1N7039CCT1<br>JANS, JANTXV, JANTX1N7039CCU1<br>JANS, JANTXV, JANTX1N7047CCT3 |

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2006-037)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil .



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